AYE
TECH HUB

• CRITICAL THINKING SERIES • NO. 05

Logical Fallacies

Arguments That Sound Right But Are Wrong

The Thinker's Field Guide to Flawed Reasoning

No. 05 of 30 | Critical Thinking Series

Awet G. Nway

Founder, AYE Tech Hub | Thinker, Educator & AI Researcher

- 20 logical fallacies — name, structure, real-world example
- The Straw Man: why it dominates political and social media debate
- The False Dilemma: how politicians limit your thinking to two choices
- Fallacies in AI output — how to spot them in generated content
- Quick-reference cheat sheet: identify any fallacy in 10 seconds

01 | What Is a Logical Fallacy?

A **logical fallacy** is an error in reasoning that makes an argument appear valid when it is actually flawed. Fallacies are dangerous precisely because they *sound convincing* — they often appeal to emotions, exploit cognitive biases, or use rhetorical tricks that bypass careful analysis. They are the primary weapons in propaganda, political manipulation, advertising, and bad-faith debate.

Knowing the name of a fallacy gives you power over it. When you can say "that's a straw man" or "that's a false dilemma", you neutralise the argument's emotional force and expose the actual weakness in the reasoning. This skill is one of the most valuable tools in both intellectual self-defence and respectful, productive disagreement.

> ***"It is not so important to be serious as it is to be serious about the important things. The monkey wears an expression of seriousness which would do credit to any great philosopher. But the monkey is serious because he itches."***
>
> — Robert Frost (on the appearance vs substance of argument)

02 | The Four Categories of Logical Fallacies

Figure 1 — The Four Categories of Logical Fallacies

Category	Core Error	How to Spot
Relevance Fallacies	The argument introduces information that has nothing to do with the actual question	Ask: "Does this information actually address the claim being made?"
Weak Induction Fallacies	The evidence is real but does not logically support the conclusion drawn from it	Ask: "Is this sample large enough? Could there be another explanation?"
Presumption Fallacies	The argument assumes something that has not been established, or limits options unfairly	Ask: "What is this argument taking for granted? Are there other options?"
Ambiguity and Distraction Fallacies	The argument uses vague language or shifts the topic to avoid the real issue	Ask: "Are the same words being used with different meanings? Is the topic changing?"

03 | The 10 Most Common Fallacies — Deep Dive

Fallacy	Structure	Real-World Example	Counter-Move
Ad Hominem (Attack the Person)	Attacks the person making the argument instead of the argument itself.	"You can't trust his climate research — he drives a petrol car."	"Even if true, that does not affect the evidence in the research."
Straw Man	Misrepresents an opponent's argument to make it easier to attack.	"She wants healthcare reform, so she wants the government to control all medicine."	"That's not what was said. The actual argument was [restate accurately]."
False Dilemma (Either/Or)	Presents only two options when more actually exist.	"You're either with us or against us."	"There are more options than just those two. For example: [list alternatives]."
Appeal to Authority	Argues something is true because an authority said so — even if irrelevant.	"This diet must work — a famous actor promotes it."	"What does the peer-reviewed evidence say, independent of who endorses it?"
Slippery Slope	Claims that one small step will inevitably lead to extreme consequences.	"If we allow same-sex marriage, next people will want to marry animals."	"Each step in a causal chain needs evidence. What proves A leads to Z?"
Hasty Generalisation	Draws a broad conclusion from too small a sample.	"I met two rude people from that country, so they're all rude."	"How large and representative was this sample? N=2 proves nothing."
Post Hoc (False Cause)	"After this, therefore because of this." Assumes correlation = causation.	"I wore my lucky shirt and we won. My shirt caused the victory."	"What controlled evidence links these events? Could it be coincidence?"
Begging the Question (Circular)	The conclusion is assumed in one of the premises.	"The Bible is true because the Bible says it is true."	"The support for the conclusion IS the conclusion. That proves nothing."
Appeal to Ignorance	Claims something is true because it has not been proved false (or vice versa).	"No one has proved ghosts don't exist, so they must be real."	"Absence of evidence is not evidence. The burden of proof lies with the claimant."
Bandwagon (Ad Populum)	Argues something is true or right because many people believe it.	"Millions of people believe in [X], so it must be true."	"Popularity does not determine truth. The earth was flat in popular belief for centuries."

The Straw Man — Why It Dominates Modern Debate

The straw man is the most common fallacy in political debate, social media, and workplace conflict because it is easy to construct and emotionally satisfying. It feels like a devastating rebuttal but actually attacks a position the other person never held. The antidote: always restate your opponent's argument accurately before responding — this is called the "Steelman" technique.

04 | The Next 10 Fallacies

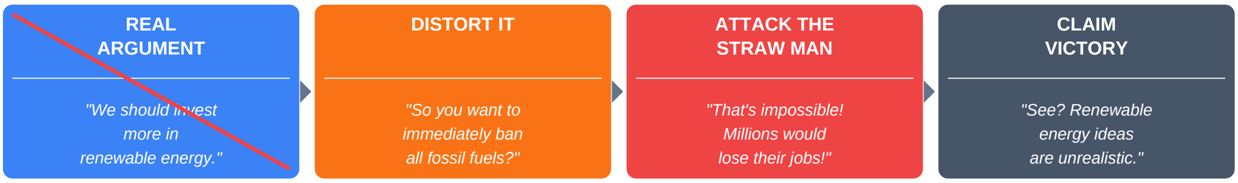
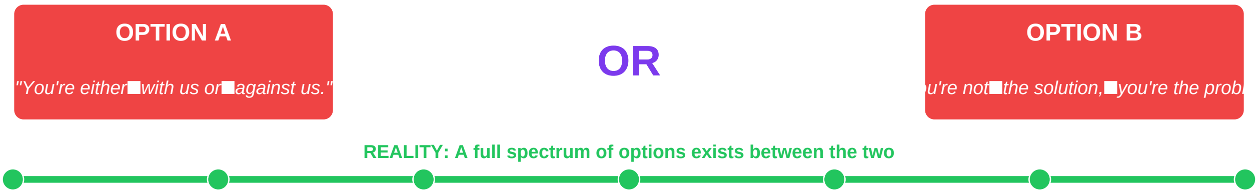


Figure 2 — The Straw Man Fallacy: Distort → Attack → Claim Victory

Figure 3 — The False Dilemma: Two Choices Presented When a Spectrum Exists



Fallacy	The Error in One Line	Quick Example
Tu Quoque ("You Too")	Deflects a criticism by pointing out that the accuser does the same thing.	"You say I lie? You've lied too!" — A's lying is not excused by B's.
Red Herring	Introduces an irrelevant topic to distract from the issue being discussed.	Politician asked about economy changes subject to national security instead.
False Equivalence	Treats two things as equal when they are significantly different in kind or degree.	"Evolution is just a theory — same as my theory that aliens built the pyramids."
Appeal to Emotion	Manipulates emotional responses instead of using logical arguments.	Using images of suffering children to support a policy with no logical connection.
Genetic Fallacy	Dismisses an argument based on its origin rather than its merit.	"That idea came from [group], so it must be wrong."
Composition/Division	Assumes what is true of the parts is true of the whole (or vice versa).	"Every player on this team is excellent, so this is an excellent team."
Equivocation	Uses the same word in two different senses to mislead.	"The sign said Fine for Parking — so I parked there." (fine = good vs fine = penalty)
Naturalistic Fallacy	Claims something is good or right simply because it is natural.	"This is a natural herb, so it must be safe and healthy."
Appeal to Tradition	Claims something is right because it has always been done that way.	"We've always done it this way — why change?"
Burden Shifting (Negative Proof)	Demands the other side disprove a claim rather than providing positive evidence.	"Prove that my homeopathic remedy does NOT work."

05 | Fallacies in AI-Generated Content

AI Behaviour	Fallacy Present	How to Spot and Challenge It
AI confidently cites a non-existent study	Appeal to Authority (fabricated)	"What is the exact citation? Let me verify it independently."

AI says "many experts believe..." with no names	Appeal to Authority + Bandwagon	"Which experts specifically? What is their evidence?"
AI presents only two options for a complex problem	False Dilemma	"What other approaches exist? Is this really a binary choice?"
AI's response begins by agreeing with you before adding nuance	Sycophancy (not a classic fallacy, but a reasoning error)	"Don't agree — challenge my assumption first. What is wrong with my premise?"
AI uses extreme example to justify a general rule	Hasty Generalisation	"Is this one case representative? What does the full evidence base show?"
AI says "this is the natural approach" or "traditionally..."	Naturalistic / Appeal to Tradition	"What is the evidence-based reason for this approach, independent of tradition?"

What comes next — CT-006

AI and Human Intelligence — Partners or Rivals? A philosophical and practical exploration of what makes human intelligence irreplaceable, where AI surpasses humans, the collaboration model, and how to position yourself to thrive in a world increasingly shaped by artificial intelligence.